



ANDERSON JUNIOR COLLEGE

2009 JC 1 H2 CHEMISTRY PROMOTIONAL EXAMINATION

Name()

PDG

CHEMISTRY

Higher 2

SECTION A: Free Response

SECTION B: Structured Questions

9647

7 Oct 2009 (Wed)

1 hour 55 min

Additional Materials: Data Booklet
 Optical Answer Sheet (OAS)

READ THESE INSTRUCTIONS FIRST

1. Write your name and PDG on all the work you hand in.
2. Answer **ALL** questions.
3. Answer:
SECTION A on writing paper
SECTION B in the spaces provided
4. You are advised **not** to exceed the duration suggested below for each section:

SECTION A: 65 min

SECTION B: 50 min

SECTION C: 50 min

AT THE END OF THE EXAMINATION

Fasten any **writing papers** at the **back** of the question paper and submit together.

Do **not** detach any page.

Section	Question	Marks
A 40 marks	1	
	2	
B 30 marks	1	
	2	
	3	
C 30 marks	1-30	
s.f / d.p		
TOTAL 100 marks		

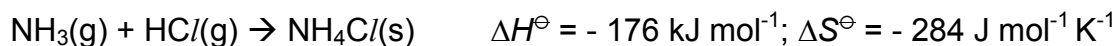
This document consists of **12** printed pages, including the cover page.

Section A – Free Response Questions

You should not spend more than 65 min on this section.

- 1 Ammonium chloride is a common chemical found in commercial cold packs. A commercial cold pack usually consists of a barrier separating the water and ammonium chloride. It is activated by breaking this barrier, allowing the chemicals to mix.

Industrial preparation of ammonium chloride involves reacting ammonia with hydrogen chloride in steel vessels coated with fibre glass or polyvinyl chloride (PVC) lining. The equation of the reaction is as follows:



- (a) (i) Suggest why the lining of the vessel is necessary.
- (ii) Construct a suitable energy cycle and use it to calculate the lattice energy of ammonium chloride using the enthalpy change of reaction for the preparation of ammonium chloride given above and the following data:

$$\Delta H_f^\ominus (\text{gaseous NH}_4^+) = +634 \text{ kJ mol}^{-1}$$

$$\Delta H_f^\ominus (\text{gaseous Cl}^-) = -243 \text{ kJ mol}^{-1}$$

$$\Delta H_f^\ominus (\text{gaseous HCl}) = -92.3 \text{ kJ mol}^{-1}$$

$$\Delta H_f^\ominus (\text{gaseous NH}_3) = -45.9 \text{ kJ mol}^{-1}$$

- (iii) Explain how you would expect the lattice energy of ammonium bromide to compare with that of ammonium chloride calculated above.

[6]

- (b) (i) Calculate the standard Gibbs free energy change for the industrial production of ammonium chloride.
- (ii) Predict and explain whether the above reaction is spontaneous at high or low temperature.
- (iii) Calculate the temperature at which the production of ammonium chloride becomes spontaneous.
- (iv) The actual temperature is lower than that calculated in (b)(iii). Explain why this is so.

[6]

(c) In an experiment to determine the effectiveness of a cold pack, a 5 g sample of ammonium chloride is dissolved in 125 g of water. Both substances are initially at 20 °C.

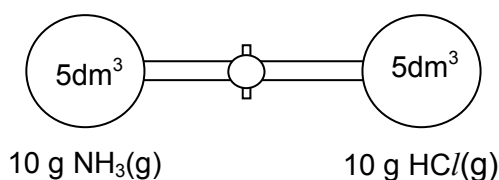
(i) By means of a chemical equation, illustrate the term *standard enthalpy change of solution of ammonium chloride*.

(ii) Given that the standard enthalpy change of solution of ammonium chloride is + 15 kJ mol⁻¹, calculate the final temperature of the solution.

[Assume that the specific heat capacity of the solution is 4.2 J K⁻¹ g⁻¹.]

[4]

(d) In a pilot trial run of the production of ammonium chloride, two 5 dm³ flasks at 25 °C are connected by a stopcock, as shown in the diagram below. One flask contains 10 g gaseous NH₃, and the other contains 10 g gaseous HCl. When the stopcock is opened, the gases react until one is completely consumed.



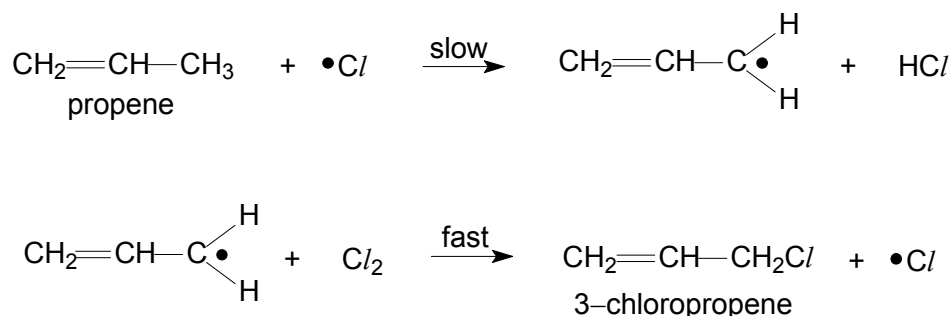
Calculate the final pressure at 25 °C after the reaction is complete.

[4]

[Total: 20]

- 2 (a) The Shell process involves the free radical substitution of gaseous propene to give 3-chloropropene at a high temperature of 400 °C.

The propagation steps of the free radical substitution are as follows:



The activation energy of the slow step is + 360 kJ mol⁻¹ and the enthalpy change of the overall propagation steps is – 167 kJ mol⁻¹.

- (i) Use the data provided to construct a reaction pathway diagram for the propagation steps.
- (ii) Identify the catalyst involved in the propagation steps of the Shell process. Explain why this is an example of homogeneous catalysis.
- (iii) Describe how the activation energy is affected by the presence of a catalyst, and explain how this increases the rate.
- (iv) What effect will the presence of the catalyst have on the rate constant for this reaction? Explain your answer.

[7]

- (b) At a lower temperature, propene can react with chlorine via a different mechanism to form an organic compound **A**. The composition of **A** by mass is C, 31.9%; H, 5.3%; Cl, 62.8%.

- (i) Use the data to calculate the empirical formula and suggest a structural formula of **A**.
- (ii) Name the type of reaction and describe the mechanism for this reaction, including curly arrows showing the movement of electrons, and all charges.

[5]

(c) Substitution reaction of 3-chloropropene with phenyllithium ($\text{C}_6\text{H}_5\text{Li}$) produces an aromatic compound **B**, C_9H_{10} .

- **B** decolourises bromine in an inert organic solvent in the absence of UV light.
- **B** reacts with hot concentrated KMnO_4 to produce a gas which gives a white precipitate with limewater.
- On reacting with steam using H_3PO_4 as catalyst, **B** gives an alcohol **C**, $\text{C}_9\text{H}_{12}\text{O}$, which consists of an equimolar (1:1) mixture of two isomers.

Heating **C** over hot Al_2O_3 produces **D**, C_9H_{10} , which is a structural isomer of **B**. Vigorous oxidation of **D** produces CH_3COOH and **E**, $\text{C}_7\text{H}_6\text{O}_2$, which reacts in NaOH(aq) to form a salt.

Deduce the structures of compounds **B**, **C**, **D** and **E** and explain the chemistry of the reactions described. [There is no need to comment on the chemistry of the formation of **B** from 3-chloropropene.]

[6]

(d) The boiling point of 3-chloropropene is 46°C while that of propene is -48°C . Explain why the boiling point of 3-chloropropene is higher than propene.

[2]

[Total: 20]

Section B - Structured Questions

You should not spend more than 50 min on this section.

Write your answers in the space provided below.

- 1 (a) Explain the terms *rate equation*, *order of reaction* and *rate constant*.

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.....[3]

- (b) You are to design an experiment to show that the overall kinetics of the reaction between magnesium metal and hydrochloric acid is first order by collecting and measuring the volume of hydrogen gas evolved.



- (i) In the space below, draw a labelled diagram of the apparatus you would use in your experiment. Your diagram should show clearly how these pieces of apparatus are assembled together.

- (ii) Give a brief outline of the experimental procedure.

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- (iii) Suggest how the data collected in the experiment can be used to show that the overall kinetics of the reaction is first order.

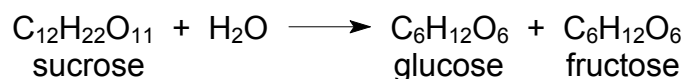
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[5]

- (c) Hydrochloric acid acts as a catalyst for the hydrolysis of sucrose. The hydrolysis can also be catalysed by the enzyme sucrase which occurs in yeast.



The results of some investigations of the rate of the hydrolysis catalysed by hydrochloric acid are shown below.

experiment	initial [HC ⁺] /mol dm ⁻³	initial [sucrose] /mol dm ⁻³	initial rate /mol dm ⁻³ s ⁻¹
1	0.10	0.10	0.024
2	0.15	0.15	0.054
3	0.20	0.10	0.048

- (i) Determine the rate equation for the hydrolysis of sucrose catalysed by hydrochloric acid. Show clearly how you obtained your answer.

- (ii) Hence, calculate the rate constant for the reaction, including units.

rate constant:

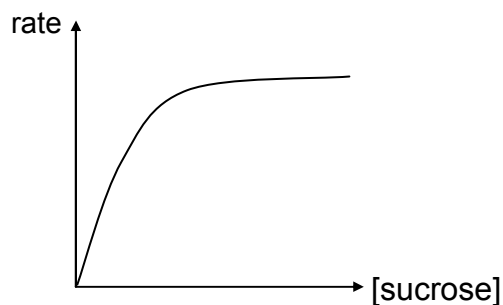
- (iii) The half-life of sucrose in experiment 1 was 3.0 s. Predict the half-life of sucrose in experiment 3, giving your reasoning.

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- (iv) The relationship between sucrose concentration and the rate of the hydrolysis catalysed by sucrase is shown in the graph below.



- I Deduce the order of the reaction at low [sucrose].

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- II Explain why the rate is independent of [sucrose] when [sucrose] is high.

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[7]

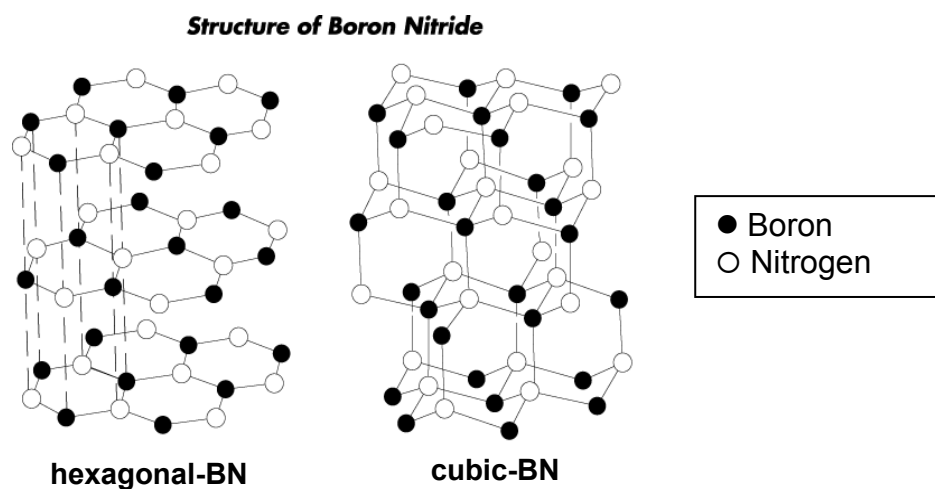
[Total: 15]

- 2 Boron nitride (BN) is a compound structurally similar to carbon lattices. It can exist in hexagonal or cubic structure.

The hexagonal-BN is a lubricant and is widely used as an additive in cosmetics such as foundations, lip-sticks and eye shadows. The cubic-BN is commonly used as an abrasive and in cutting tools.

The hexagonal-BN can be produced by reacting H_3BO_3 with NH_3 at a high temperature of 900°C .

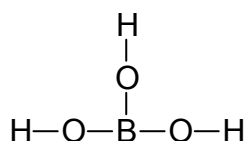
The structures of hexagonal-BN and cubic-BN are shown below.



- (a) (i) Write a balanced equation for the production of hexagonal-BN from H_3BO_3 and NH_3 .

.....

- (ii) The structural formula of H_3BO_3 is shown below. State the values of the O-B-O and B-O-H bond angles.



$\angle\text{OBO} = \dots\dots\dots$

$\angle\text{BOH} = \dots\dots\dots$

[3]

- (b) State the structure and bonding in each form of BN and explain why hexagonal-BN acts as a lubricant while cubic-BN is used as an abrasive.

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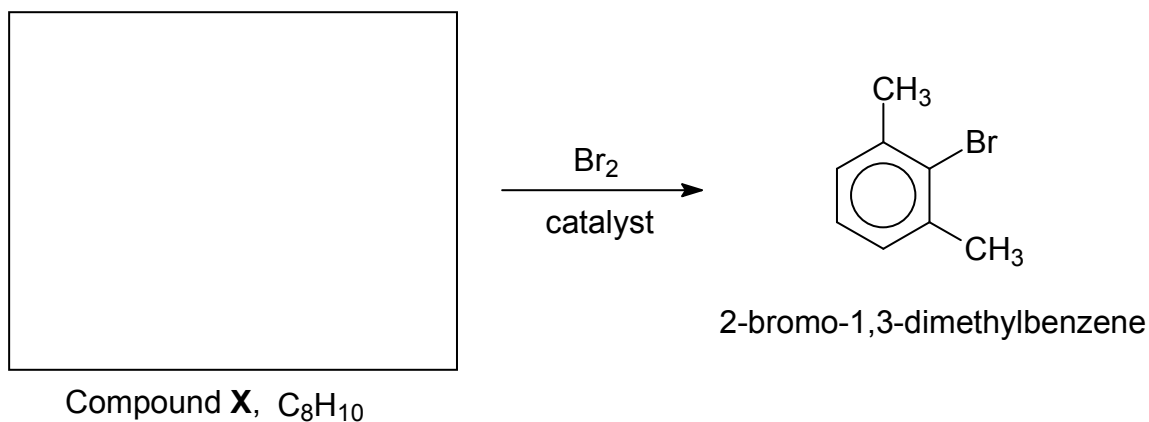
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.....[4]

[Total: 7]

- 3 Aromatic hydrocarbons are often obtained in the petrochemical industry by the process of catalytic reforming. One such hydrocarbon is compound **X**, C_8H_{10} , which reacts with bromine in the presence of a suitable catalyst to give 2-bromo-1,3-dimethylbenzene as shown below.

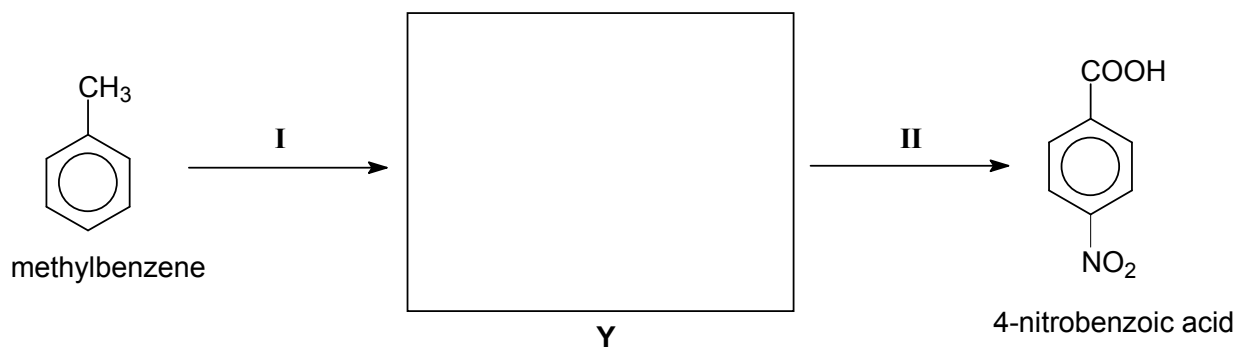


- (a) (i) Draw the full structural formula of **X** in the box above.
- (ii) Suggest a suitable catalyst for the reaction.
-
- (iii) State the type of reaction undergone by **X**.
-
- (iv) Draw the structure of the carbocation formed during the reaction and state the type(s) of hybridisation present in the structure.

Type(s) of hybridisation:

[5]

- (b) The synthesis of 4-nitrobenzoic acid from methylbenzene is a two-step reaction that can be represented as follows.



Draw the structural formula of the intermediate **Y** in the box above, and suggest the reagents used for steps **I** and **II**.

reagent for step **I**

reagent for step **II**

[3]

[Total: 8]